25 Geospatial Analysis of Storage Structures of the Mesolithic-Neolithic Transition in Britain and Ireland

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The long-term history of pit use in Great Britain and Ireland extends well beyond the Neolithic, challenging the conventional association between storage structures and early farming communities. This study therefore investigates the spatiotemporal evolution of pit technology throughout the transition from the Mesolithic to the Neolithic (approx. 8000–2400 BC cal.).

To model this technological change, we have developed a transparent and reproducible FLOSS workflow in QGIS, combining radiocarbon-dated pit contexts from the open-access XRONOS repository with a purpose-built doctoral research database that will be made available via open access. Using kernel density estimation (KDE) and inverse distance weighting (IDW) multi-temporal interpolation, we have mapped spatial changes in pit density, clustering patterns and their distribution within the landscape over time.

By analysing storage features along a long-term continuum, rather than as an isolated Neolithic phenomenon, this study offers new perspectives on the spatial organization of prehistoric resource management during the Mesolithic-to-Neolithic transition. Furthermore, it demonstrates how integrated open data sets and open-source software tools can enhance the spatial resolution of regional archaeological synthesis within a fully reproducible workflow.